

Sounderya Varagur Venugopal

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TECHNICAL SKILLS

Languages	: C, C++, Python, Julia
Robotics & Controls	: ROS / ROS 2, MoveIt 2, Gazebo, MuJoCo, Isaac Sim, Model Predictive Control, Control Systems
Machine Learning	: PyTorch, OpenCV, Reinforcement Learning, TensorRT, ONNX, Roboflow, SmolVLA
Developer Tools	: Docker, Git, CMake, GTest, CI/CD, Qt

EDUCATION

University of Maryland - College Park	MD, USA
M.Eng. Robotics (GPA 3.88)	2024 - 2026
Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram	Chennai, TN, India
B.Tech in Electronics and Communication Engineering (GPA 9.07 / 10)	2019 - 2023

WORK EXPERIENCE

Graduate Research Assistant	Feb 2025 - May 2026
University of Maryland - College Park	Onsite - College Park, MD
Collective Dynamics and Control Laboratory	
- Built a real-time bike-lane detection system by training a YOLOv8 model on a custom-annotated dataset across image-stream and point-cloud outputs. Doubled real-time inference FPS for the ReZoom perception stack by converting a YOLOv8 PyTorch model to TensorRT via ONNX export, demonstrating production-level acceleration on constrained edge hardware.	
Robotics Algorithms & Autonomous Systems (RAAS) Lab	
- Fine-tuned SmolVLA (450M-param vision-language-action model) on a Koch follower arm for tabletop manipulation tasks. - Built a mobile-manipulation pipeline on a TurtleBot2 base with a LeRobot Koch leader-follower manipulator for autonomous navigation and object pickup in structured indoor environments. - Extended ITPS (Inference-Time Policy Steering), an end-to-end UR3 manipulation pipeline encompassing automated demonstration generation in MuJoCo, diffusion policy training and real-time human steering, at inference time.	
Graduate Engineer Trainee - Mobile Robotics Division	Sep 2023 - Jun 2024
Addverb Technologies Limited	Onsite - Noida, U.P., India
Resolved a thread-synchronization defect in the core VPL stack by adding mutex locks, validated with C++ GTest, clearing site-blocking issues within 2 months. - Built a custom Visual Programming Language (VPL) interface for the in-house cobot and extended its communication capabilities via WebSockets. - Enhanced the cobot's vision pipeline with YOLOv8 gesture recognition for smoother human-robot collaboration.	
Robotics Software Intern - Mobile Robotics Division	May 2022 - Sep 2022
Addverb Technologies Limited	Onsite - Noida, U.P., India
- Interfaced mechanical and electromagnetic grippers with the Synchro collaborative arm and integrated drive-data safety flags for safe collaborative interaction. - Built a Gazebo + ROS Controllers simulation to validate the in-house robotic-arm path planners.	

PROJECTS

Autonomous Bin-Picking Robot	Source Code
- Built a fully autonomous bin-picking system across 4 object classes with no human intervention: SVM-based detection, point-cloud pose estimation (ROI filtering, statistical outlier removal, plane removal, Euclidean clustering), MoveIt 2 planning, and UR5e + Robotiq execution. - Established CI/CD via GitHub Actions with CppLint and Clang-format enforcement.	
Task-Space Motion Planner for MoveIt 2	Source Code
- Engineered a custom task-space RRT-Connect planner plugin for MoveIt 2, achieving a 100% success rate in constrained environments where standard OMPL joint-space planners completely failed. - Integrated TRAC-IK to resolve SE(3) Cartesian samples to joint configurations, boosting IK solve rate to 99.88% at 0.37 ms/solve for real-time task-space exploration. - Architected a configurable interface with a rotation-weighted distance metric that halved path lengths (1.18 m → 0.60 m) and reduced geometric deviation by 65%.	
Multi-Agent Reinforcement Learning for Drone Navigation	Source Code
- Developed custom MAPPO and IPPO controllers in PyTorch using centralized training with decentralized execution (CTDE), extending gym-pybullet-drones with hovering, spiral-formation, and leader-follower tasks. - Achieved 3× improvement over Stable-Baselines3 via centralized critics, relative coordinate encodings, and reward shaping — reducing spiral-formation RMSE to 0.25 m and collision rate to 2.1%.	
Footstep Following for Cassie Bipedal Robot	Source Code
- Designed a reference-guided RL (PPO) walking controller for the underactuated Cassie robot in MuJoCo, bridging high-level footstep planning and low-level leg actuation through a learned policy outputting PD-tracked joint targets. - Engineered a body-frame projection of the global footstep plan into an ego-centric observation space — addressing contact switching, dynamic balance, and coupled heading-velocity dynamics for sim-to-real readiness.	
RL Policy Robustness to Observation Noise (POMDP)	Source Code
- Trained a PPO cube-lift policy (Franka Panda, Isaac Lab, 1024 parallel envs) and characterized its sensitivity to observation noise across 11 levels, quantifying the MDP-to-POMDP degradation as success fell from 100% to 84% at 10 cm noise. - Showed physically bounded (3 cm) noise-augmented fine-tuning retained 97% success while unbounded augmentation caused catastrophic forgetting; implemented a 300-particle filter that cut position error 15× (7.2 to 0.47 cm), recovering >99% success as a denoising layer.	

ADDITIONAL EXPERIENCE

Medical Robotics Software Developer	Oct 2022 - May 2023
Medical Imaging, Devices, and Soft Robotics Lab, IIITDM Kancheepuram	Onsite - Chennai, TN, India
Simulated and 3D-printed a continuum-robot prototype for colonoscopy applications. - Interfaced the robot with Simulation Open Framework Architecture (SOFA) for hardware-in-the-loop testing.	
Robotics Software Developer	Jul 2020 - Jan 2022
Mars Research Station, IIITDM Kancheepuram	Onsite - Chennai, TN, India
- Worked on an interdisciplinary team designing a Mars Rover; simulated a Panda arm in Gazebo / ROS and interfaced controllers for teleoperation via MoveIt. - Built a four-wheeled mobile robot from scratch using Arduino and Raspberry Pi for teleoperation.	